SUDHANVA DESHPANDE

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EDUCATION

Purdue University West Lafayette, Indiana
Bachelor of Science Computer Engineering August 2024 - Expected May 2027

SKILLS

Programming & Developer Tools: Python, Java, C, Matlab, Agile, Linux, Git, Flask, FastAPI, VSCode, Jira, Confluence
Machine Learning: Supervised & Unsupervised Learning, RLHF, RAG, LoRA, PEFT, Sequential Models, CNNs, LSTM, Time Series Forecasting (ARIMA, Chronos), Gradient Descent, K-Fold Cross Validation, Hyperparameter Tuning (Optuna, AutoGluon), Recommender Systems, Feature Importance Analysis, K-means Clustering
LLMs & NLP: GPT, LLaMA, Gemini, Deepseek, HuggingFace Transformers (SFTTrainer, Seq2Seq), SentenceTransformers
Frameworks & Libraries: PyTorch, Tensorflow, Sci-kit Learn, Pandas, NumPy, OpenCV, DarkNet, Sktime, Matplotlib
MLOps & Cloud: VertexAI, BigQuery, Cloud Run, VPC Networks, Docker, Github Actions, CI/CD, REST APIs, Tableau

WORK EXPERIENCE

Dexcom *May 2025 - August 2025*
Software Engineering Intern

- Architected and deployed a HIPAA-hardened, ML-powered complaint correlation platform for a **leading global medical device manufacturer**, reducing risk assessment filtering from **~3 days to under 45 seconds** and accelerating patient safety decisions.
- Built a secure **FastAPI web app** on **GCP Cloud Run** with VPC networks, **Okta OAuth OIDC** authentication, and role-based access controls to safeguard PHI/PII under regulated R&D change control.
- Designed and deployed automated Jira **REST API + BigQuery** pipelines with a finetuned **two-tower Sentence-BERT recommender system**, cosine similarity search, and k-means clustering to semantically match **400K+ patient complaint**s to Jira issues, detect trends, and flag potential field actions.
- Automated deployments via **GitHub Actions CI/CD** with monitoring and rollback; presented to stakeholders up to **SVP level**, securing cross-team adoption and establishing a blueprint for future ML-driven QA tooling.

Purdue University *October 2024 - Present*
ML Researcher - *Predictive Modeling and Machine Learning Lab*

- Selected by the **Associate Dean of Research & Innovation** Dr. Lin to lead an ML initiative with Dr. Malandraki, developing ML classifiers on real patient swallow-pattern datasets to enable early Parkinson's detection.
- Tested Decision Tree, Random Forest, SVM, and KNN models, achieving **85–90% accuracy** and performing **feature importance analysis** to identify key biomarkers.
- Worked on **time series classification of EMG signals** to predict neuromuscular impairments, improving forecasting accuracy.
- Applied HuggingFace workflows for efficient model training, fine-tuning, and cross-validation; **co-authoring a paper** for submission to top tier journals.

Qoom Inc. *January 2023 - May 2025*
Software Developer Intern

- Developed and deployed an AI-powered coding assistant using **HuggingFace SFT** to generate code from natural language instructions, enabling even inexperienced users to create functional projects without manual coding.
- **Fine-tuned** the **DeepSeek Distill LLM** with **LoRA + PEFT** on large curated code datasets, applying **RLHF** and **RAG** to improve code accuracy, style consistency, and execution reliability.
- **Engineered multi-agent workflows** for automated project scoping and **integrated** the assistant directly into the Qoom platform to enhance usability and streamline project creation.

PROJECTS *August 2022 - April 2025*

Citation Contexts - *with Prof. YY Ahn at Indiana University*

- Applied large language models and ML to study whether scientific research papers are cited positively or negatively
- Converted complex citation text into interpretable numeric vectors using **sentence embedding models**
- Mapped relationships with **k-means clustering** to visualize how citations group by sentiment and thematic context.
- Identified overrepresented words and phrases in each cluster using a **log-odds ratio algorithm**, linking linguistic patterns to citation sentiment.

ASDRP Research - *Medical Imaging*

- Built a **CNN-based model** for brain tumor detection from MRI scans, achieving **98% validation accuracy** using **TensorFlow and OpenCV.**
- Applied image standardization and augmentation (rotation, scaling, noise) to reduce overfitting and improve generalization.
- **Co-authored and published** a peer-reviewed paper: *"Brain Tumor Detection Using CNN"* – Journal of Student Research.